Mamba State Space Model with Spectral Feature Conditioning for State of Health Prediction and Anomaly Detection in Lithium-Ion Batteries

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Abstract. The fast development of solar energy systems and electric vehicle applications requires robust monitoring of lithium-ion batteries, since the failure to detect capacity degradation can cause huge economic losses and safety concerns. Although the State of Health (SOH) is the most important indicator to determine the condition of the battery, accurate and real-time estimation from raw sensor data is still a big computational and methodological challenge. Conventional recurrent models struggle to learn long-term temporal dependencies in degradation cycles, and Transformer architectures are prohibitively expensive for deployment in edge environments. In this paper, we propose MambaSOH-Predictor, a lightweight framework based on the Selective State Space Model (Mamba) with physics-informed spectral feature conditioning. The architecture is available in two configurations: a high-accuracy long-sequence model ($L = 4096$) for full discharge cycles, or a compact 30-step window variant for real-time edge inference. MambaBlocks process the multivariate time-series data (voltage, current, temperature) with linear $O(L)$ complexity. Feature-wise Linear Modulation (FiLM) is used with a 57-dimensional cycle-level spectral and statistical feature vector. Uncertainty estimation is done by Monte Carlo Dropout, and anomaly detection is done by Isolation Forest in the same feature space at the same time. Combining the outputs gives us a risk matrix to prioritise predictive maintenance. The long-sequence model under a strict cross-battery protocol yields an MAE of 1.52 % and an RMSE of 1.97 % on the NASA Ames dataset, surpassing industrial thresholds. The compact configuration can complete inference in less than 100 ms on CPU, which demonstrates practical feasibility for real-time edge deployment without GPU acceleration.

Keywords: Lithium-ion batteries · State of Health (SOH) · Mamba state space model · Spectral feature conditioning · FiLM · Anomaly detection · Edge computing · Predictive maintenance.

1 Introduction

Driven by the global push for sustainable transport and decarbonisation [8], lithium-ion batteries (LIBs) have become the preferred energy-storage technology for electric vehicles (EVs) and grid-scale renewable integration. Road transport causes about 15.9 % of global greenhouse-gas emissions, mostly from passenger cars, and is a prime target for electrification [4]; EU initiatives such as “Fit for 55” and the “Electric 2035” vision further drive LIB demand, projected to grow almost sevenfold over 2022–2030 [17]. However, complex ageing mechanisms—solid-electrolyte interphase growth, active-lithium depletion and electrode-structure deterioration—cause capacity fade and internal-resistance rise, threatening the long-term reliability and safety of LIB systems [10,15].

Battery health monitoring has shifted from physics-based methods (equivalent circuit models, electrochemical simulation) to data-driven deep learning [5]. Recurrent architectures such as LSTM model degradation trajectories well but suffer from a recurrence bottleneck on long sequences [22,23]. Transformer self-attention addresses this but is quadratic, $O(L^2)$, limiting real-time deployment [13]. Mamba and selective state space models break this trade-off, matching Transformer accuracy at $O(L)$ linear complexity [6]. A key gap remains, however: most Mamba-based battery studies are unimodal, using only macro-telemetry and overlooking frequency-domain spectral features that are sensitive to the internal degradation fingerprint [1,10,12].

Moreover, existing methods mostly estimate SOH and detect anomalies separately, which is insufficient for real-time detection of anomalies from transient events or sensor faults [3], and few effective conditioning mechanisms bridge the distribution gap between high-dimensional temporal sequences and low-dimensional spectral features [10]. Building on this, we propose MambaSOH-Predictor, a selective state space model conditioned on physics-informed spectral features through Feature-wise Linear Modulation (FiLM). Amenable to edge deployment, it supports efficient, robust SOH prediction with simultaneous Isolation Forest anomaly detection for reliable real-time battery health management.

2 Related Work

2.1 Deep Sequence Modeling: From RNNs to Mamba

Section 1 traced the path from recurrent architectures to Transformers and selective state space models. This has yielded several Mamba-based battery-prognostics implementations (MambaLithium, SambaMixer and iMamba variants) that claim better accuracy and efficiency for unified SOH, SOC and RUL estimation than recurrent and attention-based baselines [7,12].

2.2 Health Indicators and Multimodal Frequency Conditioning

Health indicators range from raw telemetry (voltage/current/temperature) to more degradation-sensitive quantities such as incremental capacity and differential voltage analysis [18], electrochemical impedance spectroscopy [1], and internal gas pressure [19], enabling multi-feature rather than purely time-domain approaches [21]. Fusing such multi-domain information with high-dimensional temporal sequences is difficult because of mismatched statistical distributions, and naive concatenation commonly causes modal conflicts and suboptimal gradient flow [10]. The Hybrid Sensing Synergy Architecture (HSSA) uses Q-Former and cross-attention to align EIS features to the temporal representation space [10], whereas Feature-wise Linear Modulation (FiLM) offers a lighter alternative that adapts sequence representations directly from spectral conditioning without specialised EIS hardware [7].

2.3 Robust Diagnostics and Anomaly Detection

Designing a robust, safe AI-powered Battery Management System [3] requires point-wise state-of-health (SOH) estimation, uncertainty quantification and anomaly detection. Anomalies—from sensor noise, distributional shifts or new internal faults—accelerate degradation and are flagged by anomaly detection [3]. Hybrid methods have proven effective in practical solar-grid and EV cases, from statistical techniques (local outlier factor, Mahalanobis distance) to advanced frameworks such as Markov-constrained Isolation Forest [2]. Deploying such diagnostic tools within efficient Mamba-based architectures [10,12] marks a step toward self-aware, deployable battery health monitoring systems.

2.4 Literature Comparison Matrix

Table 1 positions this work against the four most closely related families of approaches.

Table 1. Comprehensive literature comparison matrix and structural comparison.

Study	**Method family**	**Key strength**	**Key limitation**	**Relation to this work**
LSTM / CNN-LSTM [3,23]	Sequential / hybrid CNN-RNN	Effective temporal modelling on short windows; simple implementation	Recurrence bottleneck; struggles with long-range dependencies and cross-battery generalisation	Served as primary baseline for comparison; outperformed by the proposed Mamba architecture
Transformer / PatchTST [13,11]	Attention-based Transformer	Strong global context capture via self-attention	Quadratic complexity $O(L^2)$; high memory/latency on long sequences ($L = 4096$); edge deployment challenges	Advanced baseline demonstrating limitations addressed by the linear Mamba backbone
Unimodal Mamba (Samba-Mixer, Mamba-Lithium) [12]	Selective state space model (SSM)	Linear complexity $O(L)$; efficient long-sequence modelling	Unimodal (raw telemetry only); limited physics-informed conditioning and interpretability	Core backbone adopted and extended; this work adds spectral conditioning via FiLM
HSSA (Mamba + EIS multi-modal) [10]	Mamba + Q-Former multimodal fusion	Multimodal fusion of discharge curves and impedance spectra; strong performance on NASA	Requires EIS data (not always available in the field); higher complexity for edge devices	Closely related multimodal inspiration; this work uses spectral features (FFT/statistical) as a more lightweight alternative conditioning signal

3 Methodology

3.1 Dataset and Evaluation Protocol

The present work utilises the lithium-ion battery lifetime dataset from the NASA Ames Prognostics Center of Excellence [16]. The dataset consists of 18650-type cells with a nominal capacity of 2.0 Ah cycled until the capacity falls below a safety threshold. Each discharge cycle has multivariate time-series data (voltage, current, surface temperature) and the end-of-cycle measured capacity. The State of Health at cycle k is defined as $\mathrm{SOH}(k) = 100\,\% \times Q_k/Q_{\mathrm{nominal}}$, where Q_k is the measured end-of-discharge capacity and the nominal capacity $Q_{\mathrm{nominal}} = 2.0\,\mathrm{Ah}$.

Quality filtering retained 26 of the original 34 cells, excluding B0036 (excessive noise), B0049–B0052 (incomplete sequences) and B0038–B0040 (reserved), plus cycles with missing capacity or insufficient length. Unlike prior work that splits data by time step within a cell, we use a strict cross-battery split:

- **Training set:** 23 cells;
- **Validation set:** 2 cells (B0046, B0047);
- **Testing set:** 1 cell (B0048), completely isolated from training.

All validation and test cells operate at 4 °C; to support this regime, the training set includes several 4 °C cells (B0041, B0045, B0053–B0056). Preprocessing and splitting use a fixed seed of 42. For additional robustness, a leave-one-battery-out (LOBO) protocol is applied on the window-30 variant with scalers refitted per fold (Section 4.3).

3.2 Data Preprocessing and Feature Extraction

Each cycle provides four raw channels (voltage, current, surface temperature, time stamp) plus two derived ones: the incremental-capacity ($\mathrm{d}Q/\mathrm{d}V$) curve, capturing ageing via shifts in peak amplitude and position [20], and discharge progress, the normalised accumulated discharged capacity ($[0, 1]$) giving a positional reference within the cycle. Ageing appears across the whole profile—shortened duration, shifted temperature, and a systematic decline of the $\mathrm{d}Q/\mathrm{d}V$ peak near 3.5 V—so signatures are distributed rather than confined to narrow windows, motivating the full discharge profile ($L = 4096$) as input.

All six channels are normalised to $[0, 1]$ with `MinMaxScaler` fitted only on the training set to prevent leakage. Full-cycle sequences are resampled to $L = 4096$ steps, while the compact configuration uses non-overlapping 30-step sliding windows for real-time inference (Section 3.4), with a dedicated set of six channels (four raw measurements, normalised cycle index, and estimated SOC).

Each cycle is also summarised into a 57-dimensional spectral–statistical feature vector (10 frequency-domain and 9 time-domain descriptors per channel; full list in the project repository), computed over the full cycle for optimal FFT resolution, shared across sliding windows within a cycle, and z-score normalised on the training distribution.

3.3 Architecture of MambaSOHPredictor

Core Mamba Processing in Pure PyTorch. The computational core is the Selective State Space Model (Mamba) [6], implemented in pure PyTorch without custom CUDA kernels for cross-platform operation (including Windows) and direct CPU inference (Fig. 1).

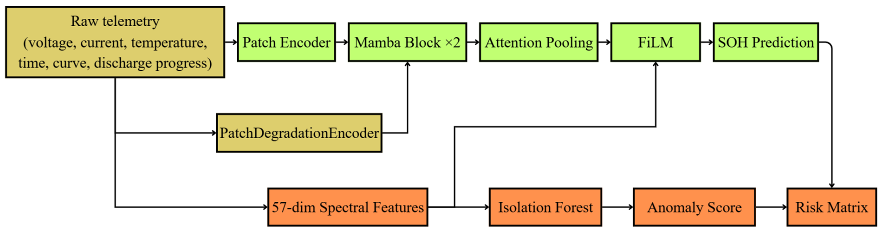

Fig. 1. Overall architecture of the proposed framework.

Each MambaBlock comprises an expansion projection ($E = 2$), a depthwise causal convolution (kernel 4), and a selective SSM scan whose state parameters are input-dependent [6], keeping or discarding information at each step—well suited to battery signals, which are mostly stationary with ageing features concentrated in transitional phases. The SSM block is thus linear, $O(L)$, versus the quadratic $O(L^2)$ of self-attention at $L = 4096$.

Patch Encoding. The sequence is split into patches of 16 steps (stride 8, 50 % overlap) and projected to about 511 tokens, as in PatchTST [11]. Each token is augmented with four local degradation statistics (RMS, peak-to-peak, standard deviation, kurtosis) via the PatchDegradationEncoder, adding dynamic context to the global spectral features.

Token Aggregation (Attention Pooling). After two MambaBlocks and LayerNorm, single-head attention pooling aggregates the 511 tokens into one cycle-level representation, learning a weight distribution over the most informative discharge phases.

Representation Modulation via FiLM. The 57-dimensional spectral vector modulates the pooled representation $\mathbf{h}$ via FiLM [14]: a two-layer MLP produces scale/shift parameters (γ, β) that transform $\mathbf{h}$ as

$$\mathbf{h} \leftarrow \big(\sigma(\gamma) + 0.5\big) \odot \mathbf{h} + \beta, \tag{1}$$

where $\sigma(\cdot)$ is the sigmoid and $\odot$ element-wise multiplication, letting the spectral summary modulate features directly rather than only at the output (Section 5).

Final Regression Head. The aggregated representation passes through a two-layer head ($64 \rightarrow 32 \rightarrow 1$, GELU, dropout 0.3) to output the continuous SOH value.

3.4 Training Setup

The long-sequence configuration uses AdamW (learning rate 5×10^{-4}, weight decay 3×10^{-4}) with a weighted SmoothL1 loss ($\beta = 0.02$, gradients clipped for residuals above 2 % SOH); near-EOL samples (SOH < 80 %) are upweighted up to 2×, decaying exponentially with distance from the threshold. A short warm-up (`CosineAnnealingWarmRestarts`, $T_0 = 3$) precedes a final stage ($T_0 = 80$, $T_{\mathrm{mult}} = 2$) whose periodic restarts escape the plateau seen under fixed decay. The compact window-30 configuration is trained separately with Adam (learning rate 5×10^{-4}, weight decay 10^{-5}), batch size 32, up to 100 epochs (patience = 15).

3.5 Uncertainty Quantification and Anomaly Detection

Uncertainty Quantification. Predictive uncertainty is estimated via Monte Carlo Dropout: 10 stochastic forward passes (dropout active at inference, batched for latency) yield a mean SOH estimate and a standard deviation as the confidence band, with negligible overhead given the compact model's 79,467 parameters.

Anomaly Detection and Risk Management. An Isolation Forest module [9] (100 estimators, contamination = 0.1, seed = 42), fitted on the 57-dimensional spectral feature space, classifies states via `decision_function` into Normal, Warning (≤ -0.1), and Anomaly (≤ -0.3), orthogonal to four SOH-derived levels: Healthy (≥ 90 %), Degrading (≥ 85 %), Maintenance Required (≥ 80 %), and End of Life (< 80 %), per NASA guidelines. A rule-based risk matrix combines these two axes with hardware safety thresholds to assign maintenance priority tiers, keeping a decoupled design: deep learning handles long-term trends, Isolation Forest detects sensor disturbances, and physical constraints act as non-overridable safeguards.

4 Experimental Results

4.1 Experimental Setup

Prediction accuracy uses MAE and RMSE in % SOH, and anomaly detection uses precision, recall and F1-score, against industrial targets MAE < 2 % and RMSE < 3 %. Models were trained in a Kaggle GPU environment in pure PyTorch (no custom CUDA kernels); latency was benchmarked on an Intel Core i7-14650HX CPU. All experiments use a fixed seed of 42 and the cross-battery protocol of Section 3.1.

4.2 Main Results

Table 2 compares four models on held-out cell B0048 (4 °C, unseen during training), all trained on the same 23/2/1 cross-battery split and seed.

Table 2. SOH prediction performance on held-out test cell B0048 (23/2/1 cross-battery split).

Model	Params	Input	MAE (%)	RMSE (%)
Naive last-SOH (oracle)†	–	Ground-truth SOH	0.89	1.49
CNN-LSTM (baseline)	61k	Telemetry, window 30	4.90	6.49
Mamba window-30 (ours)	79k	Telemetry, window 30	1.98	2.38
Mamba long $L = 4096$ (ours)	99k	Telemetry, full cycle	1.52	1.97

†Oracle reference only: it consumes the previous cycle's *measured* capacity, which real deployments cannot obtain every cycle.

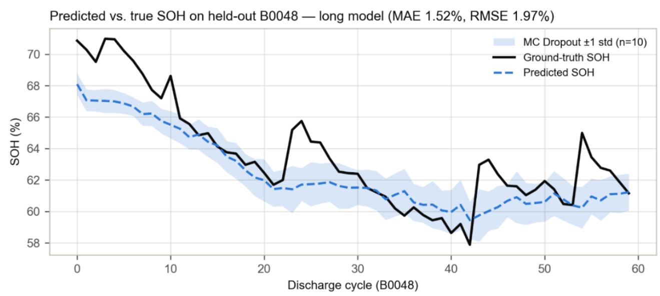

Fig. 2. Predicted vs. ground-truth SOH on B0048 – the sole fully held-out test battery under the cross-battery protocol. Shaded band: ±1 standard deviation over 10 MC Dropout samples; per-battery robustness is reported in Table 3 (LOBO).

Among telemetry-only models – the realistic deployment scenario – the long-sequence Mamba configuration achieved an MAE of 1.52 %, below the 2 % target and a 69 % relative reduction versus the CNN-LSTM baseline (4.90 %), despite the test cell being entirely unseen at an underrepresented 4 °C. The compact window-30 variant likewise cut relative error by 60 % (MAE 1.98 %), while CNN-LSTM's weak cross-battery generalisation further supports the selective state space choice.

The naive baseline (the previous cycle's measured SOH) attained an MAE of 0.89 %, outperforming all learned models, but it is not a viable competitor: it requires per-cycle capacity measurements that real systems cannot obtain, and serves only as an oracle reference for the harder, realistic task of estimating SOH from telemetry alone. As noted in Section 3.1, direct comparison with intra-cell split studies is avoided here (see Section 5 for a partial exception).

A data-centric refinement further improved the window-30 model: per-band analysis (Section 4.4) revealed a systematic negative bias in the high-SOH region under 4 °C, where training coverage reached only 67.2 % SOH. Adding one 4 °C cell (B0047, covering SOH 0–83.7 %), with no architectural or hyperparameter change, cut the MAE on B0048 from 1.98 % to 1.34 % (RMSE 2.38 % → 1.84 %; reported separately due to the modified 24/1/1 split), showing that data coverage can matter as much as architecture.

4.3 Robustness Evaluation – Leave-One-Battery-Out

A single held-out cell cannot establish statistical reliability, so we repeated the experiment via leave-one-battery-out (LOBO) on the window-30 configuration, refitting scalers per fold. Of 26 cells, 10 lacked the minimum 60 cycles, leaving 16 valid folds.

Table 3. LOBO robustness (window-30 configuration, 16 folds).

Statistic	MAE (%)	RMSE (%)
Mean ± std	3.91 ± 3.27	4.81 ± 4.17
Median	2.22	2.55
Fold B0048 (= main test battery)	1.47	1.87

Given the considerable variability, we report the full distribution rather than aggregates alone. Nine of 16 folds achieved MAE $\leq 2.3\,\%$ (best: B0042, 1.40 %); the B0048 fold (Table 3) is consistent with Table 2, confirming its selection was not favourably biased. Larger errors concentrated in cells confined to low-SOH regimes: B0045 (14.61 %, never exceeding 54 % SOH), B0033 (7.55 %) and B0054–B0056 (4.4–6.3 %), forcing extrapolation into SOH regions absent from training. LOBO thus confirms robustness in typical regimes while revealing a clear failure mode: poor extrapolation to atypical trajectories (Section 5).

4.4 Ablation and Error Analysis

Component Ablation. Table 4 quantifies each component's contribution by retraining the long-sequence model with one modification at a time, all else unchanged.

Table 4. Component ablation on test cell B0048 (long model, 23/2/1 split).

Variant	MAE (%)	RMSE (%)	Δ MAE
Full model	1.52	1.97	–
Attention pooling → last token	1.93	2.38	+0.41
d_{state} $32 \rightarrow 16$	1.24	1.61	−0.28
No EOL-weighted loss	1.32	1.69	−0.20

Attention pooling contributed most: replacing it with last-token pooling raised the MAE by 0.41 points (Table 4), the only degrading change, showing degradation information is distributed across the sequence rather than in the final token. Reducing d_{state} ($32 \rightarrow 16$) and removing the EOL-weighted loss slightly improved the MAE on B0048, but with a single test cell and few near-EOL samples these gains likely reflect cell-specific variability rather than general superiority, consistent with the LOBO spread (Section 4.3); a full LOBO ablation is left for future work. The final model keeps $d_{\text{state}} = 32$ and the EOL-weighted loss, as the larger state dimension aids long-sequence capacity and the weighted loss prioritises accuracy near the replacement threshold.

Error Analysis. The error is not uniform across SOH bands: the 80–90 % band shows the largest bias (−4.57 %), consistent with the reduced near-EOL training density.

4.5 Inference Latency

Latency was measured end-to-end via gRPC (scaler normalisation, feature extraction, batched 10-sample MC Dropout inference) over 50 calls on an Intel Core i7-14650HX CPU. The window-30 model averaged 56.1 ms (P95 = 78.3 ms), meeting the < 100 ms real-time requirement for P1 alerts; the negligible gap versus direct inference (58.7 ms) confirms minimal communication overhead. The long-sequence model trades real-time capability for accuracy, better suited to periodic offline evaluation.

4.6 Anomaly Detection

As the NASA dataset lacks explicit fault annotations, we defined a rate-based proxy label: cycles exceeding the 90th-percentile training degradation rate (0.4833 % SOH/cycle) are anomalous, matching the Isolation Forest contamination (0.1); the model (100 estimators, seed 42) was trained only on training-set spectral features.

Table 5. Anomaly detection results (Isolation Forest).

Label	Split	Threshold	Precision	Recall	F1	Positive rate
Rate	Val	Production (score $\leq -0.1 / -0.3$)	0.00	0.00	0.00	35.5 %
Rate	Val	Tuned on val (score ≤ 0.213)	0.36	1.00	0.53	35.5 %
Rate	Test	Production (score $\leq -0.1 / -0.3$)	0.00	0.00	0.00	20.6 %
Rate	Test	Tuned on val (score ≤ 0.213)	0.21	1.00	0.34	20.6 %

The production threshold failed to detect anomalies (F1 = 0); after tuning on the validation set, the test F1 improved to 0.34 (100 % recall, 0.21 precision), positioning the approach as a high-sensitivity early-warning filter requiring human verification rather than an autonomous fault detector. Since this proxy label reflects capacity degradation rather than genuine faults, future work should incorporate real IoT fault labels and semi-supervised methods (Section 5).

5 Discussion

The 1.52 % MAE under the strict cross-battery protocol shows the selective state space mechanism captures long-range degradation dependencies without the recurrence bottleneck of LSTMs or the quadratic cost of Transformers [6,10,23].

Attention pooling surpassing last-token aggregation (Section 4.4) indicates that degradation signatures are distributed rather than localised in a single terminal state, consistent with recent sequence-learning studies [10,11] and supporting our combination of patch encoding with global token aggregation.

Table 6. Comparison with recent cross-battery studies on the NASA dataset.

Work	Architecture	Params	Protocol	MAE	Latency	Robustness
SambaMixer-L (2025) [12]	MambaMixer	48.7M	10 train / 3 eval	0.51–1.20 %	Not reported	None
TIDSIT (2025) [13]	Transformer	Not reported	2 train / 1 eval	0.58 %	Not reported	None
Long model (this work)	Mamba + FiLM	99k	23 train / 2 val / 1 test	1.52 %	Not measured (see window-30)	LOBO (16 folds)
Window-30 model (this work)	Mamba + FiLM	79k	Same as above	1.98 %	56.1 ms (P95: 78.3 ms)	—

Unlike previous multimodal Mamba studies that inject electrochemical impedance spectroscopy (EIS) [10], our framework extracts FFT-based spectral descriptors directly from discharge telemetry via FiLM conditioning, avoiding specialised EIS hardware while still exploiting the higher degradation sensitivity of frequency-domain over raw time-domain signals [1,21].

The data-centric improvement in Section 4.2, where adding a single 4 °C battery reduced the MAE without any architectural change, suggests operating-condition coverage can matter more than model complexity for cross-battery generalisation, consistent with reports that prognostics performance loss stems from distribution shift rather than insufficient network capacity [10,21].

This accuracy–latency trade-off – compact for real-time edge alerts, long-sequence for periodic cloud diagnostics (Section 4.5) – mirrors recent Mamba-versus-Transformer comparisons for resource-constrained deployment [12].

As reported in Section 4.6, the low proxy-label F1 positions the Isolation Forest as a complementary early-warning module rather than an autonomous fault classifier – a limitation shared by unsupervised anomaly detection, where proxy labels cannot fully represent real operational failures [2,3].

Three limitations remain: (i) validation on the NASA dataset only (few cells, single chemistry); (ii) large LOBO variation for batteries with rare degradation trajectories; and (iii) anomaly detection on proxy rather than real fault labels. Future work should therefore validate on larger multi-chemistry datasets and real BMS fault logs, and explore semi-supervised or continual learning for robustness.

6 Conclusion

This paper presents MambaSOHPredictor, a unified Mamba-based architecture evaluated under a cross-battery, cross-temperature protocol substantially more demanding than the timestep-based splits common in the literature. The long-sequence configuration ($L = 4096$) achieved an MAE of 1.52 % and an RMSE of 1.97 % on the fully unseen, underrepresented 4 °C cell B0048, surpassing the 2 % industrial threshold and reducing the relative error by 69 % versus a CNN-LSTM baseline, while the lightweight window-30 configuration (79k parameters) kept a sub-2 % MAE at 56.1 ms mean latency (P95 = 78.3 ms) on a standard CPU.

The 16-fold leave-one-battery-out evaluation confirmed B0048 was not a favourably biased choice, while revealing a clear failure mode—degraded accuracy for cells confined to rarely observed low-SOH regimes—mitigated by expanding training coverage rather than by architectural changes. For anomaly detection, we transparently report a modest F1 of 0.34 on proxy labels, positioning the module as a sensitive early-warning filter requiring human verification rather than an autonomous classifier.

Overall, industrial-grade accuracy and real-time edge deployability are jointly achievable within a single lightweight, pure-PyTorch framework under a rigorous generalisation protocol. Future work will expand training coverage across under-represented temperature $\times$ SOH combinations, add remaining useful life (RUL) prediction, integrate SOH and anomaly outputs into a maintenance-decision risk matrix, and develop a retrieval-augmented generation (RAG) prescription layer – extensions being pursued within the broader solar battery monitoring system under development by the author group.

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